

ETC3460 Assignment 1 Part 1

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Question 1

a)

let $\text{cor} = 0.0417$

$$0.0417 = \frac{\text{Cov}(MSFT, SBUX)}{0.3 \times 0.2}$$

$$\text{Cov}(MSFT, SBUX) = 0.002502$$

Table 1: Portfolio Expected Returns and Standard Deviation

Portfolio	Expected_Return	S.Deviation
1	0.100	0.2000000
2	0.102	0.1837127
3	0.104	0.1732069
4	0.106	0.1695607
5	0.108	0.1732079
6	0.110	0.1837145
7	0.112	0.2000024
8	0.114	0.2207959
9	0.116	0.2449503
10	0.118	0.2715702
11	0.120	0.3000000

b) The Global minimum Variance (GMV) portfolio defined as the lowest variance portfolio. Based purely on answers to Part a with 11 portfolios to choose from, the GMV is given by portfolio 4 with 70% weight on SBUX and 30% weight on MSFT giving a variance of 0.0288

c)

let $\text{cor} = -1$

$$-1 = \frac{\text{Cov}(MSFT, SBUX)}{0.3 \times 0.2}$$

$$\text{Cov}(MSFT, SBUX) = -0.06$$

Table 2: Portfolio Expected Returns and Standard Deviations

Portfolio	Expected_Return	S.Deviation
1	0.100	0.20
2	0.102	0.15
3	0.104	0.10
4	0.106	0.05
5	0.108	0.00
6	0.110	0.05
7	0.112	0.10
8	0.114	0.15
9	0.116	0.20
10	0.118	0.25
11	0.120	0.30

d) The Global minimum Variance (GMV) portfolio defined as the lowest variance portfolio. Based purely on answers to Part c with 11 portfolios to choose from, the GMV is given by portfolio 5 with 60% weight on SBUX and 40% weight on MSFT giving a variance $6.9389 \cdot 10^{-18}$

e)

Note:

$$W = X_a^2$$

For the portfolio consisting of Assets MSFT and SBUX, A and B with weights x_A, x_B respectively. The portfolio variance σ_p^2 is given by: ①

$$\sigma_p^2 = x_A^2 \sigma_A^2 + x_B^2 \sigma_B^2 + 2x_A x_B \sigma_{AB}$$

where σ_A^2, σ_B^2 are the variances of Assets A and B, and σ_{AB} is the covariance between assets A, B.

Note $x_B = (1 - x_A)$, This equation ① can be rearranged:

$$\sigma_p^2 = x_A^2 \sigma_A^2 + (1 - x_A)^2 \sigma_B^2 + 2x_A(1 - x_A)\sigma_{AB}$$

$$\therefore W = \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}}, \quad \text{we want } \min_{W \in [0,1]} \sigma_p^2.$$

To find the min value of σ_p^2 we take the first derivative $\frac{d\sigma_p^2}{dw} = 2x_A \sigma_A^2 - 2(1 - x_A)\sigma_B^2 + 2(1 - x_A)\sigma_{AB} - 2x_A \sigma_{AB}$ and set the function equal to 0 to calculate its critical points.

The solution to this function with respect to x_A is: ② $x_A = \sigma_B^2 - \sigma_{AB} / (\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB})$, $1 - x_A = \sigma_A^2 - \sigma_{AB} / (\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB})$

To ensure this critical point is a min, we take the second order derivative; $\frac{d^2\sigma_p^2}{dw^2} = 2(\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB})$ which by

definition of σ_{AB} determines the equation is > 0 and therefore ② is a true minimum.

From the equation 2 we can calculate the weights of the GMV portfolio.

$$W_{MSFT} = \frac{\sigma_{SBUX}^2 - Cov}{\sigma_{MSFT}^2 + \sigma_{SBUX}^2 - 2(Cov)}$$

$$W_{MSFT} = \frac{0.04 - (-0.06)}{0.09 + 0.04 - 2(-0.06)} = 0.4$$

$$W_{SBUX} = 1 - W_{MSFT} = 1 - 0.4 = 0.6$$

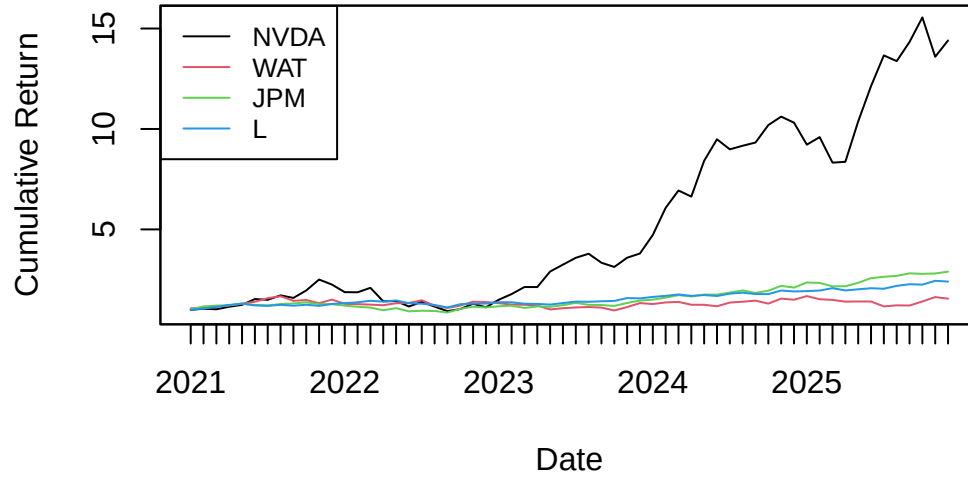
The weights for the GMV portfolio given by this equation are 40% MSFT and 60% SBUX, supporting the answer to part d.

Question 2

a)

[1] "NVDA" "WAT" "JPM" "L"

Monthly Equity Curve



b)

Table 3: Summary Statistics

	Mean	StdDev	SharpeRatio
NVDA	0.0444605	0.1428745	0.2883097
WAT	0.0073255	0.0940702	0.0431283
JPM	0.0177335	0.0680306	0.2126267
L	0.0146144	0.0485271	0.2338080

c)

Table 4: 4x4 Sample Correlation Matrix of Monthly Log Returns

	NVDA	WAT	JPM	L
NVDA	1.0000	0.0939	0.4156	0.1813
WAT	0.0939	1.0000	0.4202	0.4770
JPM	0.4156	0.4202	1.0000	0.6335
L	0.1813	0.4770	0.6335	1.0000

d) Compute GMV weight

$$\mathbf{w} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}$$

Table 5: Optimal Weights for the GMV Portfolio

Asset	Weight
NVDA	0.0487

Asset	Weight
WAT	0.0206
JPM	0.0355
L	0.8953

e)

Table 6: Portfolio Summary Statistics

Portfolio_Mean	Portfolio_StdDev	Portfolio_Sharpe
0.0160277	0.0478379	0.2667196

f) the 5% Value-at-Risk (VaR) for a one-month investment is calculated using the sample mean (μ) and standard deviation (σ) of the monthly log returns, assuming a normal distribution. The 5% quantile of returns is multiplied by the initial investment of \$50,000:

$$VaR_{5\%} = (\mu + \sigma \cdot Z_{0.05}) \times 50,000$$

Table 7: 5% Value-at-Risk (VaR) for a \$50,000 One-Month Investment

Asset	VaR_5_Percent
Nvidia (NVDA)	-9,527.36
JPMorgan Chase (JPM)	-4,708.34
Waters Corp (WAT)	-7,370.32
Loews Corp (L)	-3,260.28
GMV Portfolio	-3,132.93

g)

The Capital Asset Pricing Model (CAPM) regression is estimated using the generic expression:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t}$$

Where idiosyncratic risk is represented by the standard deviation of the residuals, $\sigma(\epsilon_i)$. The Fama-French market factors and asset returns are scaled as percentages

Table 8: CAPM Regression Output: Alpha, Beta, Risk, and Fit Metrics

Asset	Alpha	SE_Alpha	Beta	SE_Beta	Idiosyncratic_Risk	R_Squared
NVDA	2.2019	1.3292	2.2919	0.2971	10.1013	0.5064
WAT	-0.5238	1.0504	1.1515	0.2348	7.9826	0.2931
JPM	0.5847	0.6399	1.0733	0.1430	4.8627	0.4926
L	0.7019	0.5466	0.5779	0.1222	4.1538	0.2783
GMV Portfolio	0.7455	0.4866	0.6907	0.1088	3.6976	0.4101

- h) **Historical Trajectories & Cross-Asset Correlation** The monthly equity curves shown in Part a plot the past return performance of each asset. The mathematical foundation for combining these assets is established in the correlation matrix (Part c). In a real-world context, the degree to which these four specific stocks move independently dictates our ability to reduce aggregate volatility without sacrificing expected returns. For instance, the correlation between NVDA and WAT is notably low (0.0939), providing significant diversification utility within the portfolio.

Global Minimum Variance (GMV) and Downside Risk Matrix optimization (Part d) isolates the lowest-risk allocation possible for this portfolio. The weightings are heavily skewed toward the least volatile asset (L):

- **NVDA:** 4.87%, **WAT:** 2.06%, **JPM:** 3.55%, **L:** 89.53%

To quantify downside risk, the 5% Value-at-Risk (VaR) translates standard deviation into maximum expected 1-month dollar losses for a \$50,000 investment at a 95% confidence interval (Part f):

Asset	5% VaR (One-Month)
Nvidia (NVDA)	-\$9,527.36
Waters Corp (WAT)	-\$7,370.32
JPMorgan Chase (JPM)	-\$4,708.34
Loews Corp (L)	-\$3,260.28
GMV Portfolio	-\$3,132.93

The results demonstrate the practical effect of optimization: the GMV portfolio restricts extreme downside dollar risk more effectively than holding any single asset individually, even the defensive Loews Corp.

**** Asset Pricing and Systematic Risk Exposures **** The CAPM regressions isolate fundamental pricing metrics for each equity, defining how they relate to the broader market:

- **Beta (β):** Measures systematic risk. **NVDA** exhibits the highest beta at **2.2919**, indicating aggressive sensitivity to market swings. Conversely, **L** is the most defensive with a β of **0.5779**.
- **Idiosyncratic Risk & Fit (R^2):** **L** displays the lowest R^2 (**0.2783**). This low correlation to the market suggests its price action is driven predominantly by company-specific events (idiosyncratic risk) rather than macroeconomic trends.
- **Alpha (α):** Evaluating the standard errors against the estimated alphas reveals that no asset or portfolio produced a statistically significant alpha at the 5% level (e.g., NVDA $t \approx 1.65$). This indicates that historical returns were structurally consistent with the systemic market risk taken, with no evidence of abnormal “beat-the-market” performance.

Conclusion The underlying data confirms that while individual equities like NVDA provide high historical growth, they carry severe VaR parameters and extreme systematic market exposure. Implementing the weights of the GMV portfolio successfully counteracts these extremes, anchoring the allocation in defensive assets to result in a highly risk-adjusted profile that fundamentally minimizes capital drawdown.

Question 3

- a) The Fama-French three factor (FF3F) regression model over the following subperiods is as follows:

January 1997 to December 2011

The fitted Fama-French three-factor regression for the first subperiod is reported below:

Table 10: FF3F Regression Coefficients: Jan 1997 - Dec 2011

	Estimate	Std. Error	t value	Pr(> t)
Intercept (Alpha)	0.2745	0.1855	1.4796	0.1408
Mkt-RF (Beta)	0.8506	0.0386	22.0178	0.0000
SMB	-0.0533	0.0532	-1.0029	0.3173
HML	0.2824	0.0570	4.9577	0.0000

January 2012 to December 2025

The fitted Fama-French three-factor regression for the second subperiod is reported below:

Table 11: FF3F Regression Coefficients: Jan 2012 - Dec 2025

	Estimate	Std. Error	t value	Pr(> t)
Intercept (Alpha)	-0.3032	0.1275	-2.3786	0.0185
Mkt-RF (Beta)	0.9280	0.0307	30.1954	0.0000
SMB	-0.0060	0.0495	-0.1214	0.9036
HML	0.4335	0.0361	11.9949	0.0000

- b) From the results of the regression, Period 1 defines $\beta_1 = 0.85056$, that is to say 1% increase in the market's excess return, the Manufacturing portfolio's excess return increases by 0.85056%. For period 2 this is 0.928025%, indicating during period 2, the manuf portfolio is more sensitive to market movements in Period 2.

To test whether this portfolio tracks the market at a 5% significance level through the hypothesis test,

$$H_0 : \beta_1 = 1 \quad H_1 : \beta_1 \neq 1$$

Sub-Periods 1 and 2, denoted by t_1 and t_2 respectively;

$$t_1 = \frac{0.85056 - 1}{0.03863} = -3.868, \quad t_2 = \frac{0.928025 - 1}{0.030734} = -2.342$$

The degrees of freedom are 176 and 164 respectively so The Critical value for t_1 is ± 1.9735 and t_2 is ± 1.9745

Therefore for both sub periods 1 and 2 since $|t| > t_{Critical}$ we reject the null, and conclude at a 5% significance level we the beta IS statistically different from 1, therefore the Manufacturing portfolio does not perfectly track the market from Jan 1997 to Dec 2011, nor from Jan 2012 to Dec 2025

- c)

Table 12: CAPM Regression Coefficients: Jan 1997 - Dec 2011

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.3489	0.1987	1.7560	0.0808
ff3.p1\$mkt_excess	0.8085	0.0403	20.0684	0.0000

Table 13: CAPM Regression Coefficients: Jan 2012 - Dec 2025

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.3232	0.1720	-1.8790	0.062
ff3.p2\$mkt_excess	0.9316	0.0399	23.3479	0.000

Period 1 Test:

To test for the validity of the CAPM over the FF3F model using the J-test at a 5% significance level:

$$H_0 : \beta_{SMB} = \beta_{HML} = 0$$

$$H_1 : \text{at least one is non-zero}$$

Under CAPM:

$$2.659 = \sqrt{\frac{RSS_0}{178}}$$

$$RSS_0 = 1258.51$$

and under French Fama 3 factor:

$$2.465 = \sqrt{\frac{RSS_1}{176}}$$

$$RSS_1 = 1069.4156$$

Hence J-Test:

$$J = \frac{RSS_0 - RSS_1}{RSS_1 / (T - K - 1)}$$

$$J = \frac{1258.51 - 1069.4156}{1069.4156 / 176}$$

$$J = 31.120$$

Period 2 Test:

To test for the validity of the CAPM over the FF3F model using the J-test at a 5% significance level:

$$H_0 : \beta_{SMB} = \beta_{HML} = 0$$

$$H_1 : \text{at least one is non-zero}$$

Under CAPM:

$$2.152 = \sqrt{\frac{RSS_0}{166}}$$
$$RSS_0 = 768.7633$$

and under French Fama 3 factor:

$$1.579 = \sqrt{\frac{RSS_1}{164}}$$
$$RSS_1 = 408.8915$$

Hence J-Test:

$$J = \frac{RSS_0 - RSS_1}{RSS_1/(T - K - 1)}$$
$$J = \frac{768.7633 - 408.8915}{408.8915/164}$$
$$J = 144.339$$

At a 5% level of significance, the critical value for a Chi-squared distribution with 2 degrees of freedom is 5.991. The computed J-Test statistics for period 1 is 31.120 and for Period 2, 144.339. Both are greater than the Critical value therefore we reject the null hypothesis ($H_0 : \beta_{SMB} = \beta_{HML} = 0$) in both sub periods. That is we have sufficient evidence to suggest that the Fama-French 3-Factor does not reduce to CAPM during either the 1997–2011 and 2011-2025 sub periods.

d)

The client's claim is fundamentally flawed due to two primary analytical errors: ignoring statistical significance and relying on a misspecified model.

Whilst the Client correctly points out the CAPM intercept, or alpha for the 1997-2011 is positive, (0.34895), the validity of this fails when evaluated using a t-test to show if this value is statistically significant compared to 0. At a 5% significance level, the p-value of 0.0808 > 0.05, therefore we would fail to reject the null hypothesis that alpha is 0, ($H_0 : \alpha = 0$). The observed deviation from zero is within the expected range of random noise, meaning there is no evidence of true excess returns over the market risk taken from the portfolio.

Additionally the client bases their buy signal from the alpha derived from a purely CAPM model. However From the tests conducted earlier, for the 1997-2011 sub period, we definitely rejected the null hypothesis that the Fama-French 3-Factor (FF3F) model reduces to the CAPM. (see Section 3 part C)

This proves CAPM regression calculated is an inadequate formulation. If we evaluate the portfolio's performance however using the FF3F model for period 1, $\alpha = 0.27445$, with a p-value of 0.141. Since this p-value is again greater than 0.05 we cannot conclude the portfolio generated statistically significant abnormal returns.

Recommendation: Do not buy based on the positive estimated alpha-risk.

The decision to initiate a buy cannot be statistically supported at a 5% significance level from the regression models of CAPM or FF3F results, however this does not indicate whether the trade is good or bad, as it cannot be proved or disproved if buying will produce significant alpha over excess market returns.

Appendix: R Code

```
MFST.mew <- 0.12
SBUX.mew <- 0.10

MSFT.sigma <- 0.3
SBUX.sigma <- 0.2
# 1. Variable Assignments
cov.cov <- 0.002502

p1.mew <- SBUX.mew
p1.sigma <- SBUX.sigma
p2.mew <- (0.1 * MFST.mew) + (0.9 * SBUX.mew)
p2.sigma <- sqrt((0.1^2 * MSFT.sigma^2) + (0.9^2 * SBUX.sigma^2) + 2*(0.1*0.9 * cov.cov))
p3.mew <- (0.2 * MFST.mew) + (0.8 * SBUX.mew)
p3.sigma <- sqrt((0.2^2 * MSFT.sigma^2) + (0.8^2 * SBUX.sigma^2) + 2*(0.2*0.8 * cov.cov))
p4.mew <- (0.3 * MFST.mew) + (0.7 * SBUX.mew)
p4.sigma <- sqrt((0.3^2 * MSFT.sigma^2) + (0.7^2 * SBUX.sigma^2) + 2*(0.3*0.7 * cov.cov))
p5.mew <- (0.4 * MFST.mew) + (0.6 * SBUX.mew)
p5.sigma <- sqrt((0.4^2 * MSFT.sigma^2) + (0.6^2 * SBUX.sigma^2) + 2*(0.4*0.6 * cov.cov))
p6.mew <- (0.5 * MFST.mew) + (0.5 * SBUX.mew)
p6.sigma <- sqrt((0.5^2 * MSFT.sigma^2) + (0.5^2 * SBUX.sigma^2) + 2*(0.5*0.5 * cov.cov))
p7.mew <- (0.6 * MFST.mew) + (0.4 * SBUX.mew)
p7.sigma <- sqrt((0.6^2 * MSFT.sigma^2) + (0.4^2 * SBUX.sigma^2) + 2*(0.6*0.4 * cov.cov))
p8.mew <- (0.7 * MFST.mew) + (0.3 * SBUX.mew)
p8.sigma <- sqrt((0.7^2 * MSFT.sigma^2) + (0.3^2 * SBUX.sigma^2) + 2*(0.7*0.3 * cov.cov))
p9.mew <- (0.8 * MFST.mew) + (0.2 * SBUX.mew)
p9.sigma <- sqrt((0.8^2 * MSFT.sigma^2) + (0.2^2 * SBUX.sigma^2) + 2*(0.2*0.8 * cov.cov))
p10.mew <- (0.9 * MFST.mew) + (0.1 * SBUX.mew)
p10.sigma <- sqrt((0.9^2 * MSFT.sigma^2) + (0.1^2 * SBUX.sigma^2) + 2*(0.9*0.1 * cov.cov))
p11.mew <- MFST.mew
p11.sigma <- MSFT.sigma

# 2. Data Aggregation
portfolio_results <- data.frame(
  Portfolio = 1:11,
  Expected_Return = c(p1.mew, p2.mew, p3.mew, p4.mew, p5.mew, p6.mew, p7.mew, p8.mew, p9.mew, p10.mew, p11.mew),
  S.Deviation = c(p1.sigma, p2.sigma, p3.sigma, p4.sigma, p5.sigma, p6.sigma, p7.sigma, p8.sigma, p9.sigma, p10.sigma, p11.sigma)
)

# 3. Table Rendering
knitr::kable(portfolio_results, caption = "Portfolio Expected Returns and Standard Deviation")
# 1. Variable Assignments (Execution occurs silently)
cov.cov <- -0.06

p1.mew <- SBUX.mew
p1.sigma <- SBUX.sigma
p2.mew <- (0.1 * MFST.mew) + (0.9 * SBUX.mew)
p2.sigma <- sqrt((0.1^2 * MSFT.sigma^2) + (0.9^2 * SBUX.sigma^2) + 2*(0.1*0.9 * cov.cov))
```

```

p3.mew <- (0.2 * MFST.mew) + (0.8 * SBUX.mew)
p3.sigma <- sqrt((0.2^2 * MSFT.sigma^2) + (0.8^2 * SBUX.sigma^2) + 2*(0.2*0.8 * cov.cov))
p4.mew <- (0.3 * MFST.mew) + (0.7 * SBUX.mew)
p4.sigma <- sqrt((0.3^2 * MSFT.sigma^2) + (0.7^2 * SBUX.sigma^2) + 2*(0.3*0.7 * cov.cov))
p5.mew <- (0.4 * MFST.mew) + (0.6 * SBUX.mew)
p5.sigma <- sqrt((0.4^2 * MSFT.sigma^2) + (0.6^2 * SBUX.sigma^2) + 2*(0.4*0.6 * cov.cov))
p6.mew <- (0.5 * MFST.mew) + (0.5 * SBUX.mew)
p6.sigma <- sqrt((0.5^2 * MSFT.sigma^2) + (0.5^2 * SBUX.sigma^2) + 2*(0.5*0.5 * cov.cov))
p7.mew <- (0.6 * MFST.mew) + (0.4 * SBUX.mew)
p7.sigma <- sqrt((0.6^2 * MSFT.sigma^2) + (0.4^2 * SBUX.sigma^2) + 2*(0.6*0.4 * cov.cov))
p8.mew <- (0.7 * MFST.mew) + (0.3 * SBUX.mew)
p8.sigma <- sqrt((0.7^2 * MSFT.sigma^2) + (0.3^2 * SBUX.sigma^2) + 2*(0.7*0.3 * cov.cov))
p9.mew <- (0.8 * MFST.mew) + (0.2 * SBUX.mew)
p9.sigma <- sqrt((0.8^2 * MSFT.sigma^2) + (0.2^2 * SBUX.sigma^2) + 2*(0.2*0.8 * cov.cov))
p10.mew <- (0.9 * MFST.mew) + (0.1 * SBUX.mew)
p10.sigma <- sqrt((0.9^2 * MSFT.sigma^2) + (0.1^2 * SBUX.sigma^2) + 2*(0.9*0.1 * cov.cov))
p11.mew <- MFST.mew
p11.sigma <- MSFT.sigma

# 2. Data Aggregation
# Compiling the isolated variables into a structured data frame
portfolio_results <- data.frame(
  Portfolio = 1:11,
  Expected_Return = c(p1.mew, p2.mew, p3.mew, p4.mew, p5.mew, p6.mew, p7.mew, p8.mew, p9.mew, p10.mew, p11.mew),
  S.Deviation = c(p1.sigma, p2.sigma, p3.sigma, p4.sigma, p5.sigma, p6.sigma, p7.sigma, p8.sigma, p9.sigma, p10.sigma, p11.sigma)
)

# 3. Table Rendering
knitr::kable(portfolio_results, caption = "Portfolio Expected Returns and Standard Deviations")
MSFT.w <- (SBUX.sigma^2 - cov.cov)/(MSFT.sigma^2 + SBUX.sigma^2 - 2*cov.cov)
SBUX.w <- (1 - MSFT.w)
library("quantmod")
options("getSymbols.warning4.0"=FALSE)
getSymbols(Symbols=c("NVDA", "WAT", "JPM", "L"), from="2020-12-01", to="2025-12-31",
  auto.assign=TRUE, warnings=FALSE)

NVDA <- NVDA$NVDA.Adjusted
WAT <- WAT$WAT.Adjusted
JPM <- JPM$JPM.Adjusted
L <- L$L.Adjusted

prices <- merge(NVDA, WAT, JPM, L)

prices.M <- to.monthly(prices, OHLC=FALSE)
colnames(prices.M) <- c("NVDA", "WAT", "JPM", "L")

prices.M.ret <- prices.M / lag(prices.M, 1) - 1
prices.M.ret <- na.omit(prices.M.ret)
equity.curve.M <- cumprod(1 + prices.M.ret)

```

```

# Plotting Execution
plot.zoo(equity.curve.M,
        screens = 1,
        col = 1:4,
        xlab = "Date",
        ylab = "Cumulative Return",
        main = "Monthly Equity Curve")

legend("topleft",
      legend = colnames(equity.curve.M),
      col = 1:4,
      lty = 1,
      cex = 0.8)

# Monthly Log Returns
NVDA.M.l.ret <- na.omit(diff(log(prices.M$NVDA)))
WAT.M.l.ret  <- na.omit(diff(log(prices.M$WAT)))
JPM.M.l.ret  <- na.omit(diff(log(prices.M$JPM)))
L.M.l.ret    <- na.omit(diff(log(prices.M$L)))
# Merge
log.returns.all <- merge(NVDA.M.l.ret, WAT.M.l.ret, JPM.M.l.ret, L.M.l.ret)
colnames(log.returns.all) <- c("NVDA", "WAT", "JPM", "L")
# 2. Define the risk-free rate
rf_annual <- 0.04
rf_monthly_log <- log(1 + rf_annual) / 12
# Compute Mean, Standard deviation and Sharpe Ratio
mean_returns <- apply(log.returns.all, 2, mean)
sd_returns <- apply(log.returns.all, 2, sd)
sharpe_ratio <- (mean_returns - rf_monthly_log) / sd_returns
stats_summary <- data.frame(
  Mean = mean_returns,
  StdDev = sd_returns,
  SharpeRatio = sharpe_ratio
)

knitr::kable(stats_summary, caption = "Summary Statistics")
# Compute covariance and correlation
cov.mat <- cov(log.returns.all)
cor.mat <- cor(log.returns.all)

# Render the correlation matrix natively as a formatted table
knitr::kable(cor.mat, digits = 4, caption = "4x4 Sample Correlation Matrix of Monthly Log Return")
# 1. Computation
ones <- rep(1, ncol(cov.mat))
inv.cov.mat <- solve(cov.mat)
w.gmv.portfolio <- (inv.cov.mat %*% ones) / as.numeric(t(ones) %*% inv.cov.mat %*% ones)

# 2. Data Aggregation
# Extracting the numeric values and mapping them to their respective asset names
gmv_weights_df <- data.frame(

```

```

  Asset = c("NVDA", "WAT", "JPM", "L"),
  Weight = as.numeric(w.gmv.portfolio)
)

# 3. Table Rendering
knitr::kable(gmv_weights_df, digits = 4, caption = "Optimal Weights for the GMV Portfolio")
# Define the risk-free rate
rf_annual <- 0.04
rf_monthly_log <- log(1 + rf_annual) / 12
# Compute Mean, Standard deviation and Sharpe Ratio
asset_means <- apply(log.returns.all, 2, mean)
port_mean_return <- as.numeric(t(w.gmv.portfolio) %*% asset_means)
port_variance <- as.numeric(t(w.gmv.portfolio) %*% cov.mat %*% w.gmv.portfolio)
port_sd <- sqrt(port_variance)

port_sharpe_ratio <- (port_mean_return - rf_monthly_log) / port_sd
stats_summary_port <- data.frame(
  Portfolio_Mean = port_mean_return,
  Portfolio_StdDev = port_sd,
  Portfolio_Sharpe = port_sharpe_ratio
)

knitr::kable(stats_summary_port, caption = "Portfolio Summary Statistics")
initial_investment <- 50000

# NVDA
NVDA.M.l.ret <- na.omit(NVDA.M.l.ret)
NVDA.xbar <- mean(NVDA.M.l.ret)
NVDA.sd <- sd(NVDA.M.l.ret)
NVDA.q.5 <- NVDA.xbar + NVDA.sd * qnorm(0.05)
VaR.5.NVDA <- NVDA.q.5 * initial_investment

# JPM
JPM.M.l.ret <- na.omit(JPM.M.l.ret)
JPM.xbar <- mean(JPM.M.l.ret)
JPM.sd <- sd(JPM.M.l.ret)
JPM.q.5 <- JPM.xbar + JPM.sd * qnorm(0.05)
VaR.5.JPM <- JPM.q.5 * initial_investment

# WAT
WAT.M.l.ret <- na.omit(WAT.M.l.ret)
WAT.xbar <- mean(WAT.M.l.ret)
WAT.sd <- sd(WAT.M.l.ret)
WAT.q.5 <- WAT.xbar + WAT.sd * qnorm(0.05)
VaR.5.WAT <- WAT.q.5 * initial_investment

# L
L.M.l.ret <- na.omit(L.M.l.ret)
L.xbar <- mean(L.M.l.ret)

```

```

L.sd <- sd(L.M.1.ret)
L.q.5 <- L.xbar + L.sd * qnorm(0.05)
VaR.5.L <- L.q.5 * initial_investment

# GMV Portfolio
GMVp.xbar <- port_mean_return
GMVp.sd <- port_sd
GMVp.q.5 <- GMVp.xbar + GMVp.sd * qnorm(0.05)
VaR.5.GMVp <- GMVp.q.5 * initial_investment

# Data Aggregation
var_results <- data.frame(
  Asset = c("Nvidia (NVDA)", "JPMorgan Chase (JPM)", "Waters Corp (WAT)", "Loews Corp (L)", "GMV Portfolio"),
  VaR_5_Percent = c(VaR.5.NVDA, VaR.5.JPM, VaR.5.WAT, VaR.5.L, VaR.5.GMVp)
)

# Table Rendering
knitr::kable(var_results,
  digits = 2,
  format.args = list(big.mark = ","),
  caption = "5% Value-at-Risk (VaR) for a $50,000 One-Month Investment")
library(frenchdata)
library(xts)

# 1. Data Retrieval and Formatting
ff3 <- download_french_data("Fama/French 3 Factors")
ff3_monthly_raw <- ff3$subsets$data[[1]]
ff3_monthly_raw$date <- as.yearmon(as.character(ff3_monthly_raw$date), format = "%Y%m")

ff3_xts <- xts(
  ff3_monthly_raw[, c("Mkt-RF", "SMB", "HML", "RF")],
  order.by = as.Date(ff3_monthly_raw$date)
)
colnames(ff3_xts) <- c("mkt_excess", "smb", "hml", "rf")
index(ff3_xts) <- as.yearmon(index(ff3_xts))

# Aligning period and scaling returns
ff3.MRK <- ff3_xts["2021-01-01/2025-12-31"]
lr.stock <- log.returns.all * 100

# 2. CAPM Regressions
capm.NVDA <- lm(lr.stock$NVDA ~ ff3.MRK$rf ~ ff3.MRK$mkt_excess)
capm.WAT <- lm(lr.stock$WAT ~ ff3.MRK$rf ~ ff3.MRK$mkt_excess)
capm.JPM <- lm(lr.stock$JPM ~ ff3.MRK$rf ~ ff3.MRK$mkt_excess)
capm.L <- lm(lr.stock$L ~ ff3.MRK$rf ~ ff3.MRK$mkt_excess)

# GMV Portfolio Regression
w <- w.gmv.portfolio
gmv.ret.xts <- lr.stock[, "NVDA"]*w[1] + lr.stock[, "WAT"]*w[2] + lr.stock[, "JPM"]*w[3] + lr.stock[, "L"]*w[4]

```

```

colnames(gmv.ret.xts) <- c("gmw_ret")
capm.GMV <- lm(gmv.ret.xts - ff3.MRK$rf ~ ff3.MRK$mkt_excess)

# 3. Data Extraction and Aggregation
extract_metrics <- function(model, asset_name) {
  mod_summary <- summary(model)
  data.frame(
    Asset = asset_name,
    Alpha = mod_summary$coefficients[1, 1],
    SE_Alpha = mod_summary$coefficients[1, 2],
    Beta = mod_summary$coefficients[2, 1],
    SE_Beta = mod_summary$coefficients[2, 2],
    Idiosyncratic_Risk = mod_summary$sigma,
    R_Squared = mod_summary$r.squared
  )
}

capm_results_df <- rbind(
  extract_metrics(capm.NVDA, "NVDA"),
  extract_metrics(capm.WAT, "WAT"),
  extract_metrics(capm.JPM, "JPM"),
  extract_metrics(capm.L, "L"),
  extract_metrics(capm.GMV, "GMV Portfolio")
)

# 4. Table Rendering
knitr::kable(capm_results_df,
  digits = 4,
  caption = "CAPM Regression Output: Alpha, Beta, Risk, and Fit Metrics")

# 1. Data Retrieval and Preparation
ind5 <- download_french_data("5 Industry Portfolios")
id5.df <- ind5$subsets$data[[1]]
id5.df$date <- as.yearmon(as.character(id5.df$date), format = "%Y%m")

id5.xts <- xts(
  id5.df[, c("Cnsmr", "Manuf", "HiTec", "Hlth", "Other")],
  order.by = as.Date(id5.df$date)
)
index(id5.xts) <- as.yearmon(index(id5.xts))

id5.manuf <- id5.xts$Manuf
index(ff3_xts) <- as.yearmon(index(ff3_xts))

# 2. Period 1: January 1997 to December 2011
id5.manuf.period1 <- id5.manuf["1997-01/2011-12"]
ff3.p1 <- ff3_xts["1997-01/2011-12"]

# FF3F Regression 1
ff3.fit1 <- lm(id5.manuf.period1 - ff3.p1$rf ~ ff3.p1$mkt_excess + ff3.p1$smb + ff3.p1$hml)

```

```

# 3. Period 2: January 2012 to December 2025
id5.manuf.period2 <- id5.manuf["2012-01/2025-12"]
ff3.p2 <- ff3_xts["2012-01/2025-12"]

# FF3F Regression 2
ff3.fit2 <- lm(id5.manuf.period2 ~ ff3.p2$rf ~ ff3.p2$mkt_excess + ff3.p2$smb + ff3.p2$hml)
# Extract coefficients and rename rows for clarity
coef_p1 <- summary(ff3.fit1)$coefficients
rownames(coef_p1) <- c("Intercept (Alpha)", "Mkt-RF (Beta)", "SMB", "HML")

knitr::kable(coef_p1,
              digits = 4,
              caption = "FF3F Regression Coefficients: Jan 1997 - Dec 2011")
# Extract coefficients and rename rows for clarity
coef_p2 <- summary(ff3.fit2)$coefficients
rownames(coef_p2) <- c("Intercept (Alpha)", "Mkt-RF (Beta)", "SMB", "HML")

knitr::kable(coef_p2,
              digits = 4,
              caption = "FF3F Regression Coefficients: Jan 2012 - Dec 2025")
# Restricted Regression Period 1&2:
capm_restricted.fit1 <- lm(id5.manuf.period1-ff3.p1$rf ~ ff3.p1$mkt_excess)
knitr::kable(summary(capm_restricted.fit1)$coefficients, digits = 4, caption = "CAPM Regression")

capm_restricted.fit2 <- lm(id5.manuf.period2-ff3.p2$rf ~ ff3.p2$mkt_excess)
knitr::kable(summary(capm_restricted.fit2)$coefficients, digits = 4, caption = "CAPM Regression")

```